



Electronic I

Lecture 6

Regulator and Diode-Based Logic Circuits

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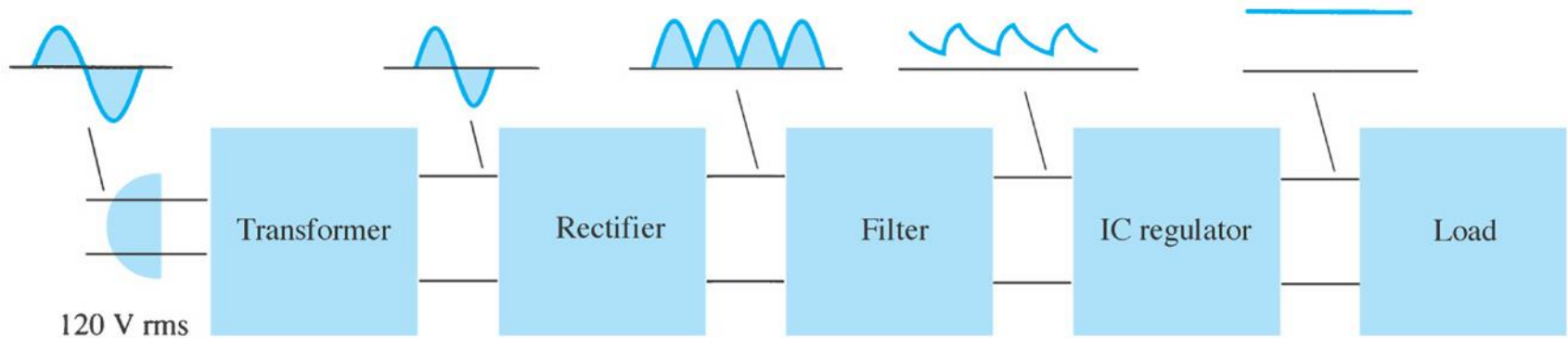
First Year / Spring Semester

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Introduction

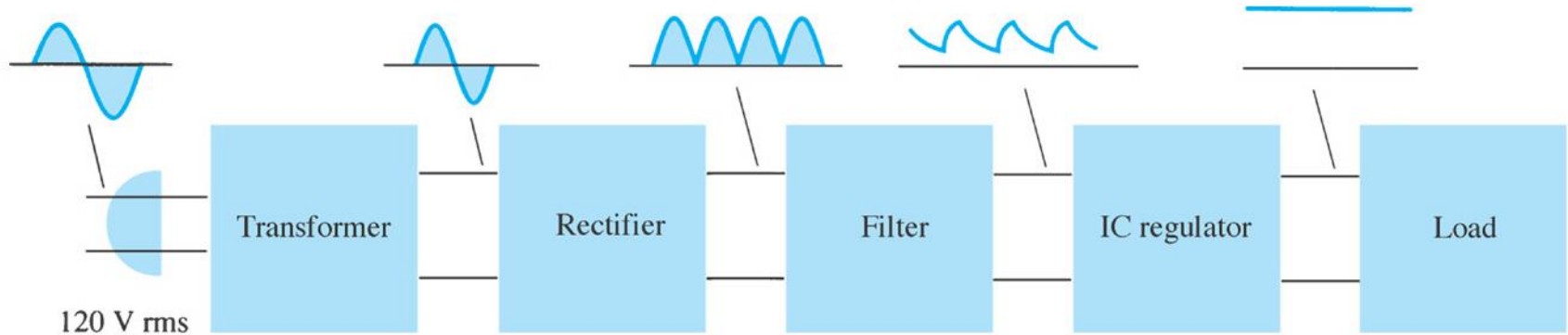
- Batteries are often shown on a schematic diagram as the source of DC voltage but usually the actual DC voltage source is a power supply.
- There are many types of power supply. Most are designed to convert high voltage AC mains electricity to a suitable low voltage supply for electronics circuits and other devices.
- A more reliable method of obtaining DC power is to transform, rectify, filter and regulate an AC line voltage.
- A power supply can be broken down into a series of blocks, each of which performs a particular function.

Introduction



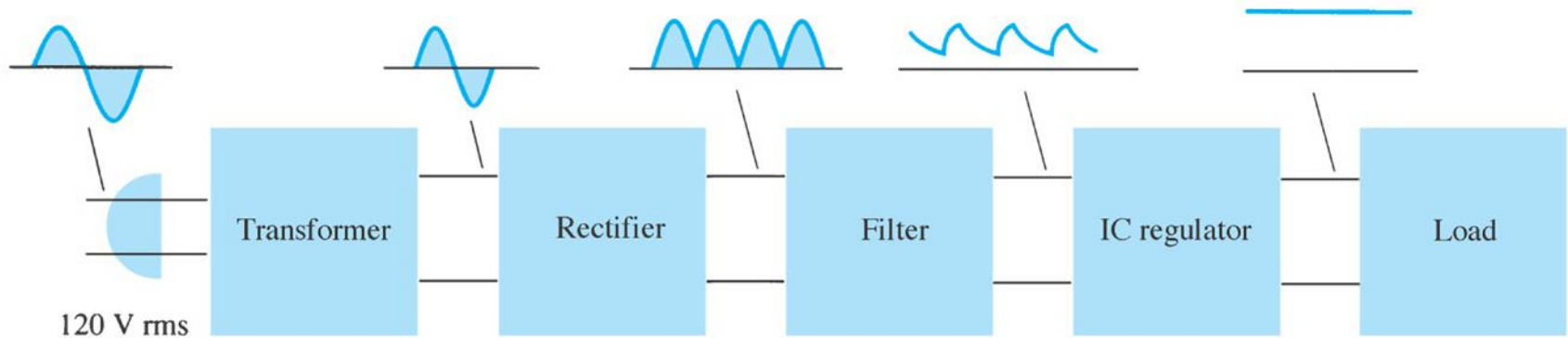
- ***Power supply***: a group of circuits that convert the ***standard ac voltage*** (120 V, 60 Hz) provided by the wall outlet to ***constant dc voltage***
- ***Transformer***: a device that step up or step down the ***ac voltage*** provided by the wall outlet to a desired amplitude through the *action* of a *magnetic field*

Introduction



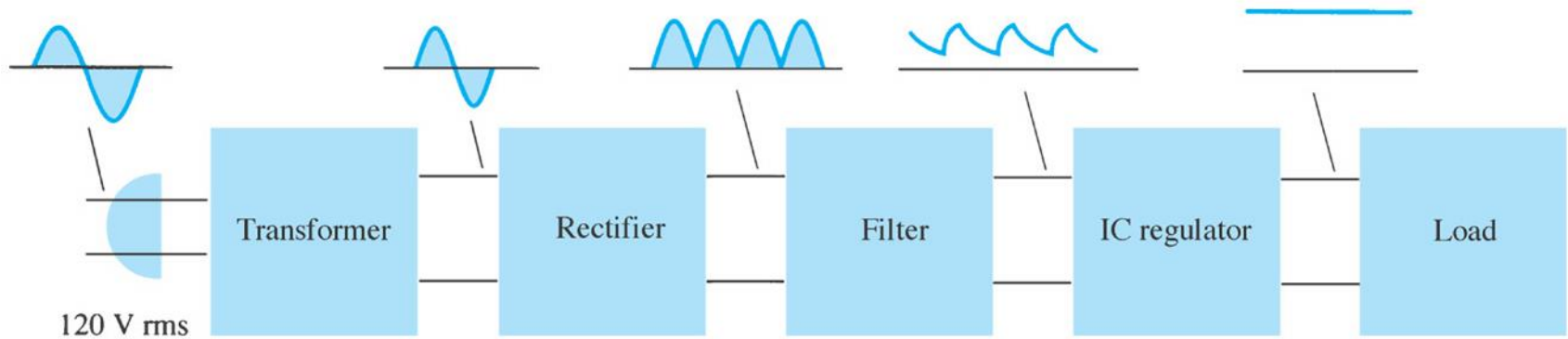
- ***Rectifier***: a diode circuit that converts the ***ac input voltage*** to a ***pulsating dc voltage***
- The pulsating dc voltage is **only suitable** to be used as a battery charger, but **not good enough** to be used as a dc power supply in a radio, stereo system, computer and so on.

Introduction



- There are two basic types of rectifier circuits:
 - Half-wave rectifier
 - Full-wave rectifier - Center-tapped & Bridge full-wave rectifier
- In summary, a full-wave rectified signal has **less ripple** than a half-wave rectified signal and is thus better to apply to a filter.

Introduction

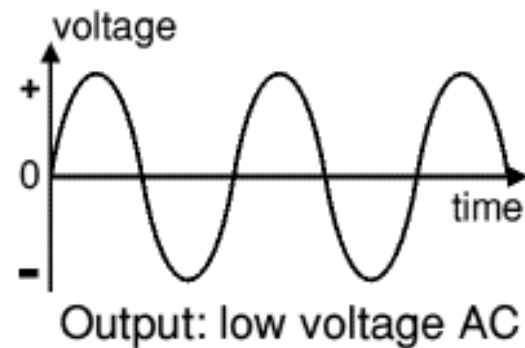
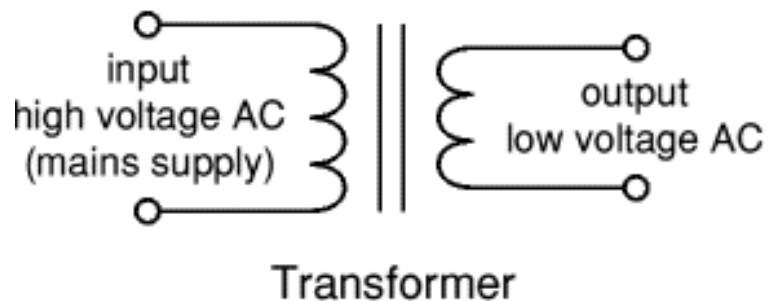


- ***Filter***: a circuit used to reduce the fluctuation in the rectified output voltage or ripple. This provides a ***steadier*** dc voltage.
- ***Regulator***: a circuit used to produces a ***constant*** dc output voltage by reducing the ripple to negligible amount. One part of power supply.

Power Supplies

➤ Transformer

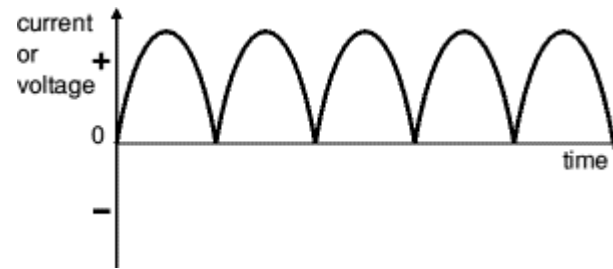
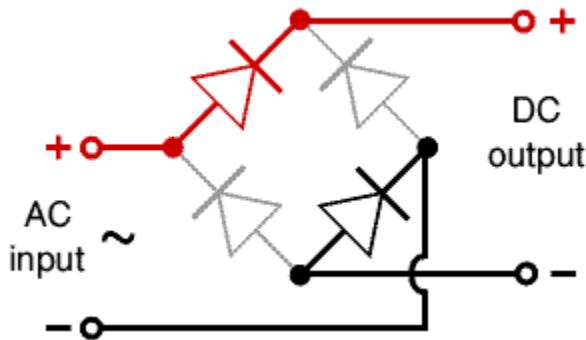
- Transformers convert AC electricity from one voltage to another with little loss of power.
- Transformers work only with AC and this is one of the reasons why mains electricity is AC.
- Step-up transformers increase voltage.
- Step-down transformers reduce voltage.



Power Supplies

➤ Rectifier

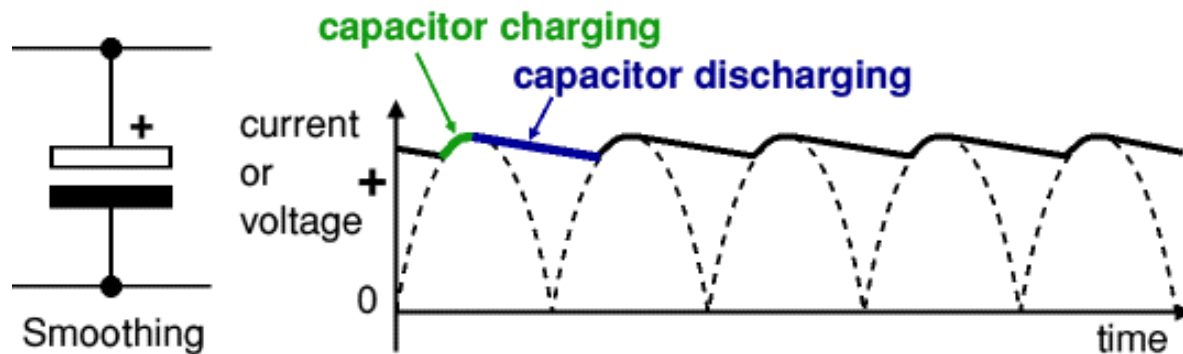
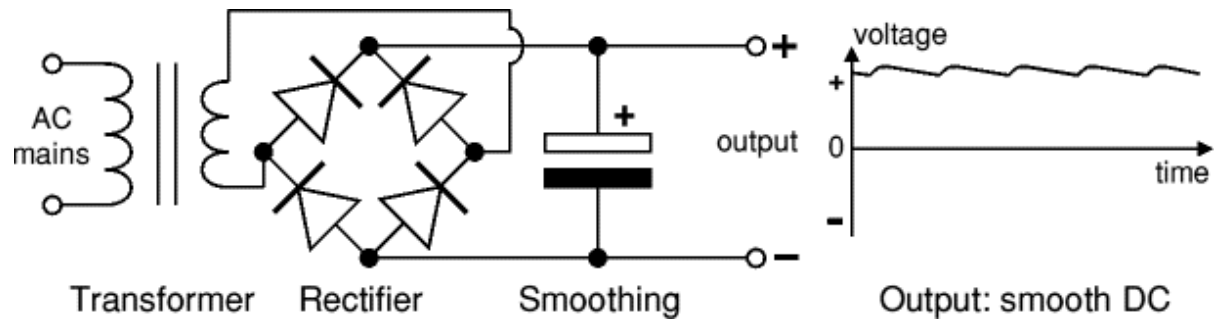
Produces full-wave varying DC.



A bridge rectifier can be made using four individual diodes, but it is also available in special packages containing the four diodes required.

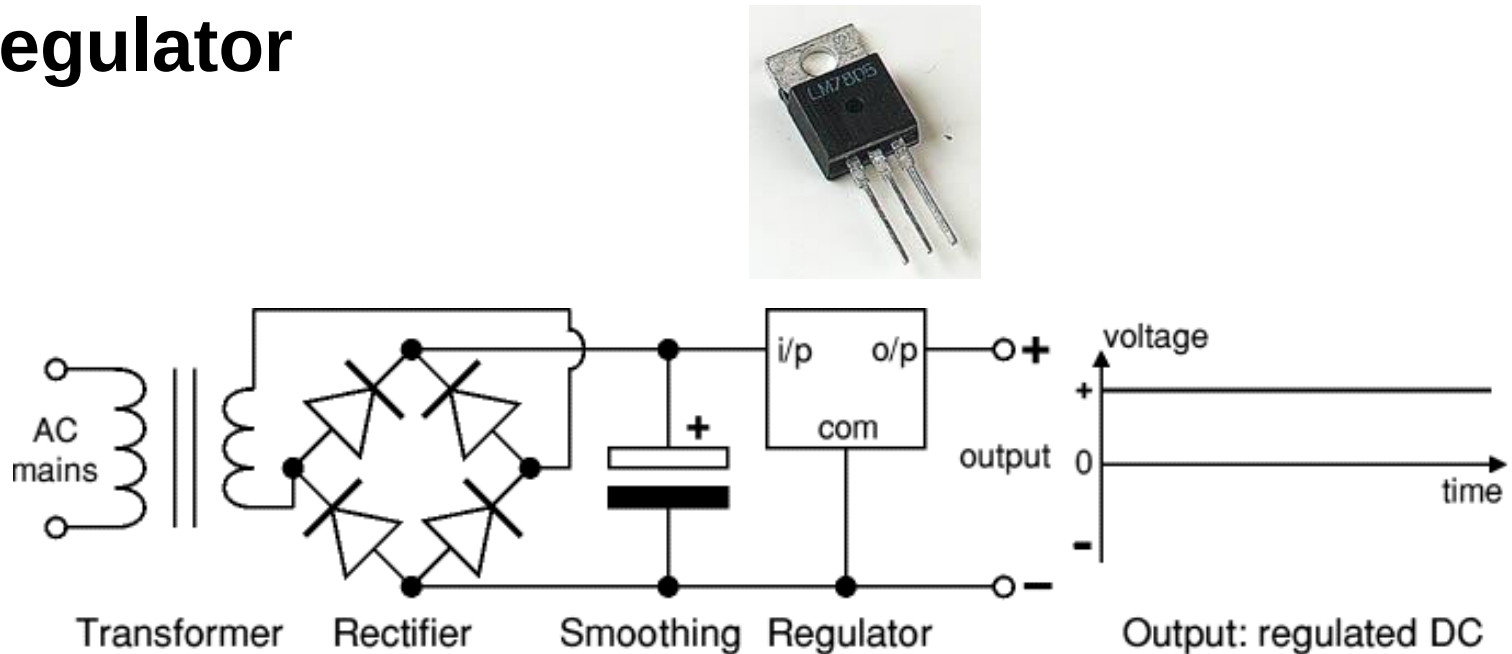
Power Supplies

➤ Smoothing



Power Supplies

➤ Regulator



Voltage regulator ICs are available with fixed (typically 5, 12 and 15V) or variable output voltages.

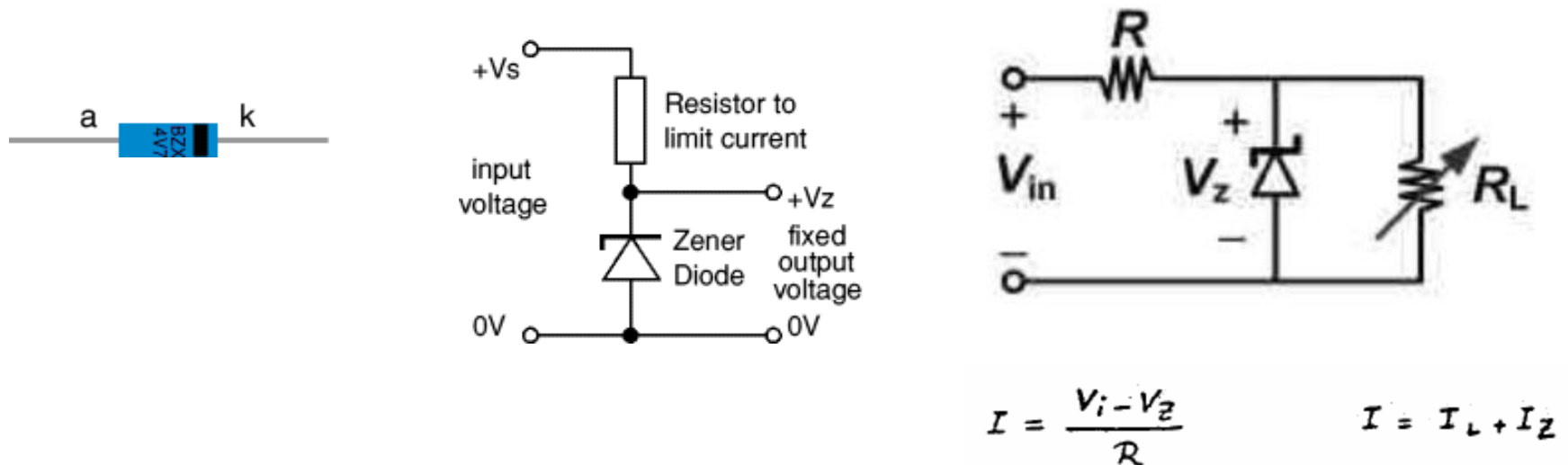
Zener diode regulator

➤ Regulator

- Zener diode regulator

For low current power supplies - a simple voltage regulator can be made with a resistor and a zener diode connected in reverse as shown in the diagram.

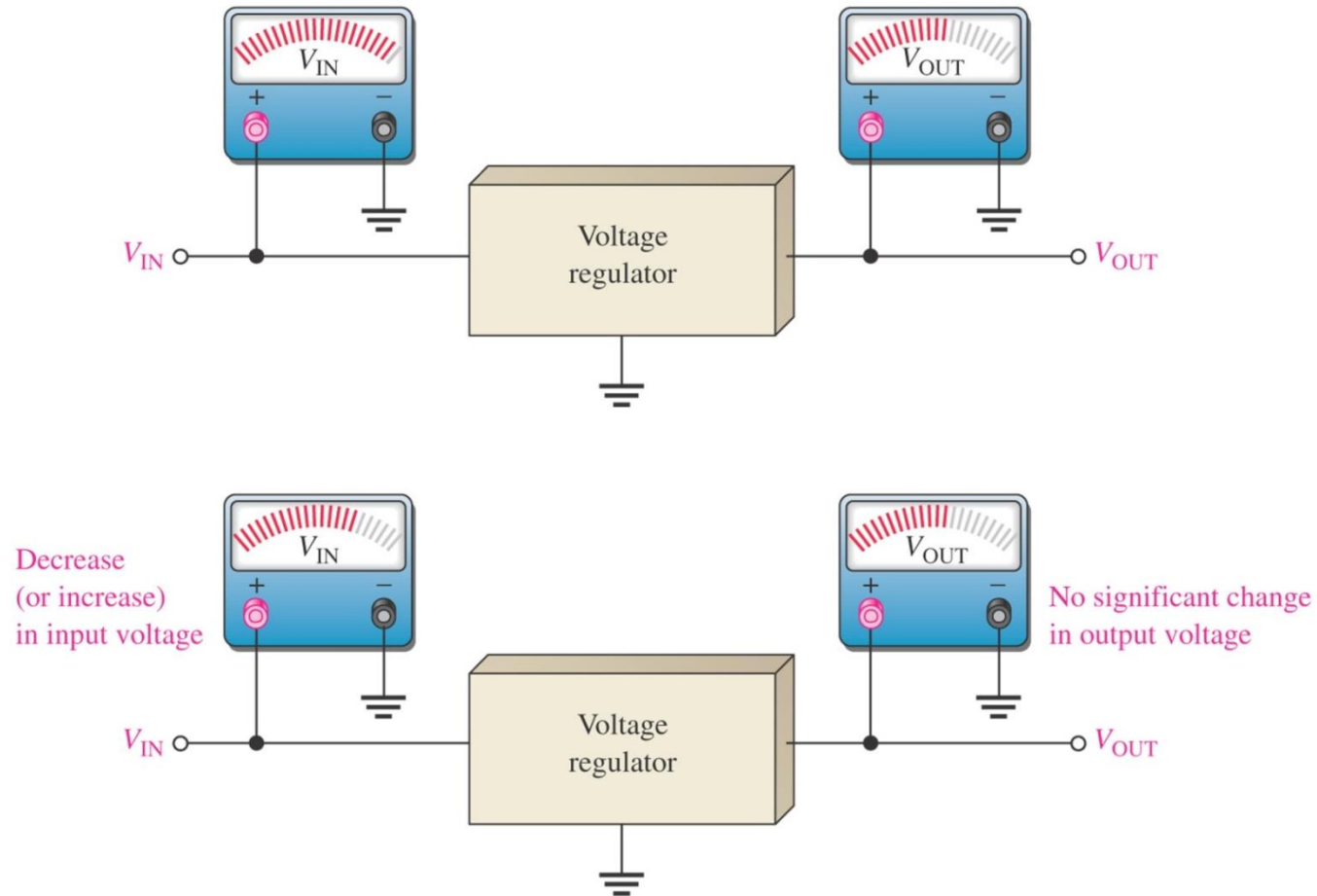
Zener diodes are rated by their breakdown voltage V_Z and maximum power P_Z (typically 400mW or 1.3W).



Voltage Regulation

- Two basic categories of voltage regulation are:
 - ❑ line regulation
 - ❑ load regulation
- The purpose of **line regulation** is to maintain a nearly constant output voltage when the **input voltage** varies.
- The purpose of **load regulation** is to maintain a nearly constant output voltage when the **load** varies

Line Regulation



Line regulation: A change in input (line) voltage does not significantly affect the output voltage of a regulator (within certain limits)

Line Regulation

- Line regulation can be defined as the percentage change in the output voltage for a given change in the input voltage.

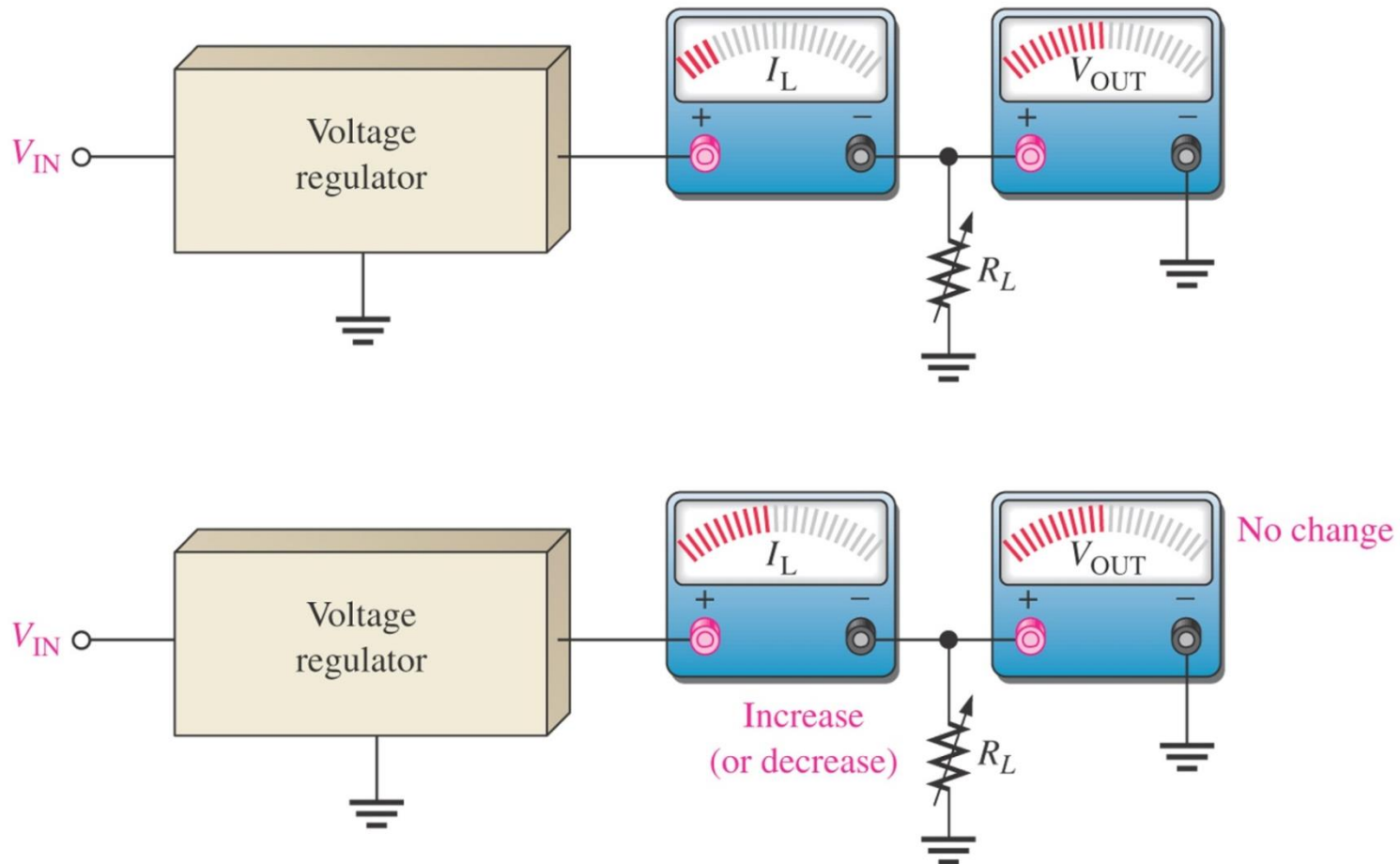
$$\text{Line regulation} = \left(\frac{\Delta V_{OUT}}{\Delta V_{IN}} \right) \times 100\%$$

Δ means “a change in”

- Line regulation can be calculated using the following formula:

$$\text{Line regulation} = \frac{(\Delta V_{OUT} / V_{OUT}) \times 100\%}{\Delta V_{IN}}$$

Load Regulation



Load regulation: A change in load current (due to a varying R_L) has practically no effect on the output voltage of a regulator (within certain limits)

Load Regulation

- Load regulation can be defined as the percentage change in the output voltage from no-load (NL) to full-load (FL).

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}} \right) \times 100\%$$

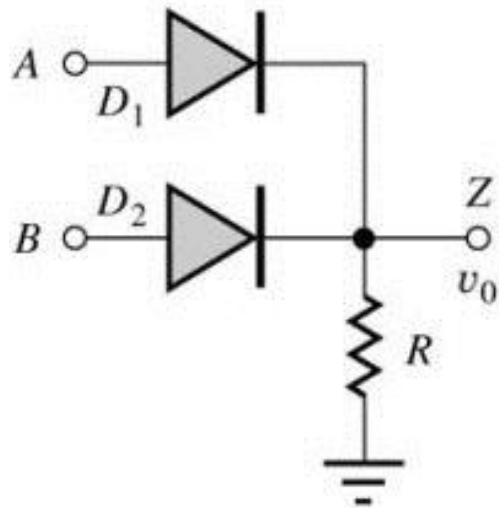
- Where:

V_{NL} = the no-load output voltage

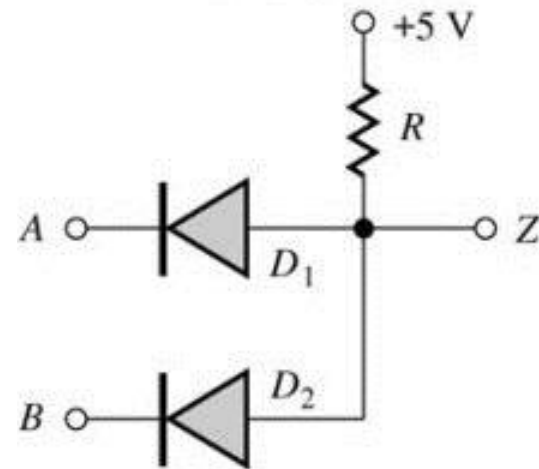
V_{FL} = the full-load output voltage

Diode-Based Logic Circuits

- Diodes can with resistive loads to implement simple logic gates



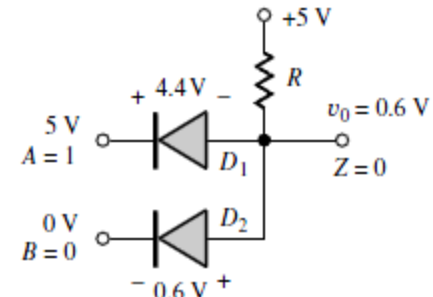
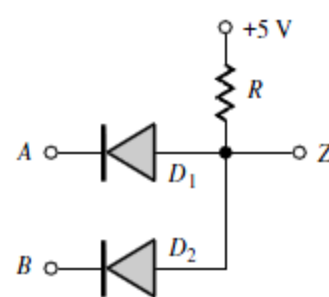
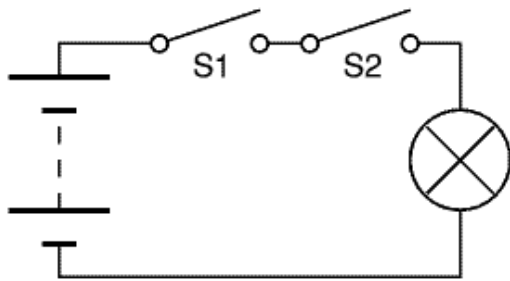
Diode OR gate



Diode AND gate

Logic Gates: AND

The simplest gates are AND and OR. They can be built from switches or using the simplest form of electronic logic - diode logic.



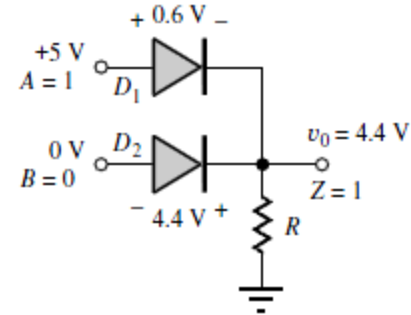
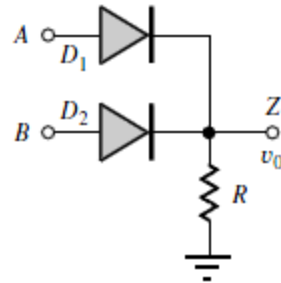
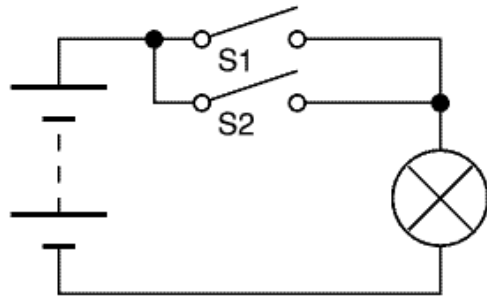
$A = 0, B = 0 \rightarrow$ both diodes are forward biased $\rightarrow$ both diodes conduct $\rightarrow$ out is LOW $\rightarrow 0$.

$A = 0, B = 1 \rightarrow$ DB is reverse biased $\rightarrow$ does not conduct,
DA is forward biased $\rightarrow$ conducts $\rightarrow$ out is LOW $\rightarrow 0$.

$A = 1, B = 0 \rightarrow$ DA is reverse biased $\rightarrow$ does not conduct,
DB is forward biased $\rightarrow$ conducts $\rightarrow$ out is LOW $\rightarrow 0$.

$A = 1, B = 1 \rightarrow$ both diodes are reverse biased $\rightarrow$
both the diodes do not conduct $\rightarrow$ out is HIGH $\rightarrow 1$.

Logic Gates: OR



$A = 0$, $B = 0 \rightarrow$ both diodes are reverse biased $\rightarrow$ does not conduct $\rightarrow$ out is LOW $\rightarrow 0$.

$A = 0$, $B = 1 \rightarrow$ D_A is reverse biased $\rightarrow$ does not conduct,
 D_B is forward biased $\rightarrow$ conducts $\rightarrow$ out is HIGH $\rightarrow 1$.

$A = 1$, $B = 0 \rightarrow$ D_B is reverse biased $\rightarrow$ does not conduct,
 D_A is forward biased $\rightarrow$ conducts $\rightarrow$ out is HIGH $\rightarrow 1$.

$A = 1$, $B = 1 \rightarrow$ both diodes are reverse biased $\rightarrow$
both the diodes conduct $\rightarrow$ out is HIGH $\rightarrow 1$.

Diode Transistor Logic

Since diode gates are limited to AND and OR functions, the diodes can be combined with transistors to complete the basic logic functions such as the following NAND gate

